

Enhanced Stock Market Forecasting Using Novel LSTM with Improved Feature Engineering

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Abstract— Financial forecasting in stock markets is a complicated activity as stock market data is volatile and largely non-linear. The conventional machine learning approaches, i.e. Support Vector Machines (SVM) and Random Forests (RF) can hardly embrace the temporal dependencies, whereas traditional Long Short-Term Memory (LSTM) models tend to perform poorly in the case of high volatility and inappropriate selection of features. This paper introduces a new volatility-sensitive LSTM architecture, which combines all the hybrid financial indicators (i.e., momentum, volatility, and trend-based engineered features) and an adaptive training approach based on dynamic learning rate control and dropout-based regularization. The suggested model can well describe the short-term variations in stock movement as well as long-term dependencies. TCS.NS stock data experimental estimates indicate that the modified LSTM yields the lowest error rates (MAE = 328.65, RMSE = 364.11) and a positive $R^2 = 0.43$, compared to the RF, SVM, and classical LSTM models. The distinctiveness of the work is determined by its feature-adaptive structure and convergence-stable design of learning, which allows predicting the market in dynamic situations effectively. The suggested solution modifies predictive stability and accuracy besides aiding in the progression of dependable, clarifiable, and scalable decision support frameworks to the area of financial forecasting uses.

Keywords— Stock Market Prediction, Long Short-Term Memory (LSTM), Machine Learning, Support Vector Machine (SVM), Random Forest (RF), Time-Series Forecasting, Deep Learning, Financial Data Analysis, Optimization etc.

I. INTRODUCTION

Stock market is central to world economies since it offers investment and wealth creation environment as well as financial decision making [1]. Stock price movement however is complex in nature and thus is very volatile, dynamic, and non-linear. The problem solving on predicting stock price is very difficult to be agreed upon by the researcher and the investors, for the reason that many elements involved in it, ranging from economic indicators, investor sentiment, politics and world events etc. [2].

Over the years, many computational techniques have been adopted in attempts to model and predict the stock market behaviors. The difference between the traditional statistical methods to capture the market data and the non-linear dependencies of market data as observed by the non-statisticians, is that the traditional methods do not work well with non-linear dependencies [3,4]. Even new generations of machine learning algorithms (e.g. Support Vector Machines (SVM) and Random Forests (RF)) are promising to be more precise in prediction, however they do not take into account the time series sequential relations of financial information [5]. A more recent usage of it involves deep learning models, and more specifically, Long Short-Term Memory (LSTM) [6] networks, which have drawn a lot of attention due to their ability to represent long-range temporal dependence and intricate data patterns [7]. Nonetheless, these developments

also present a number of issues with conventional LSTM models, such as their susceptibility towards hyperparameter choice, their issues with feature engineering and their general lack of resiliency in sudden market volatility [8]. This necessitates the urgent requirement of better architectures which would be able to increase the predictive performance but not compromise generalization on different sets of data and various market situations [9].

The latest developments in the field of deep learning and specifically the Long Short-Term Memory (LSTM) architecture have shown impressive results in learning long-term temporal dependence in financial time series. Nevertheless, even the traditional LSTMs have serious problems, such as being susceptible to high volatility, sensitive to hyperparameter configurations, and unable to capture the complex nonlinear interactions of features endogenous to stock market data.

To address these, the proposed study will present a volatility sensitive, feature added LSTM model that incorporates hybrid, technical features (trend-based, momentum-based, and volatility-based) with adaptive training system that will stabilize convergence and strengthening when the market is volatile. The proposed architecture uses feature fusion and adaptive learning rate unlike traditional LSTM models where the only input is the raw price sequence and can thus better represent both short and long-term dependency in noisy financial markets.

The comparison between Support Vector Machine (SVM), Random Forest (RF), and standard LSTM models on TCS.NS stock data shows that the proposed model has the lowest error values (MAE = 328.65, RMSE = 364.11) and the best generalization ($R^2 = 0.43$). These findings affirm that the proposed changes help enhance predictive stability and flexibility in unpredictable markets to a big extent.

All the major contributions of this research are summed up as follows:

- The construction of a new LSTM-based design that will combine feature level fusion with new adaptive learning features to deal with volatility.
- The overall comparative analysis of the model performance and classical ML and DL benchmark under different market conditions.
- Delivery of a strong, understandable, and scalable decision support framework to investors, traders and analysts that work in volatile financial settings.

Therefore, the paper contributes to the science of financial forecasting by seeking to solve critical issues of volatility, sparsity of features, and training instability, and provides an effective, predictable platform to the actual stock market forecasting. The rest of the paper is organized as follows Section II Literature review and motivation (Stock prediction. III Proposed Modified LSTM Model: Methodology and

Architect III-A The Methodology and Architecture of the Proposed Modified LSTM Model. In Section IV, paper present multi-base model mathematical modeling. The results and implications are discussed in Section V, and thus, conclude the paper with future work in the last section VI. Intra class variations are prevalent i.e. plant disease data. K-Nearest Neighbors (KNN) classifier was also another most widely used classifier in this respect due to its simplicity and nonparametric behavior. If the disease symptoms such have an obvious visual prediction, KNN will get good accuracies, but it was very sensitive to the noise, and was also depended heavily on the distance metric used and the computation is difficult for large data sets. Decision Trees and their variations, as the ensemble version Random Forests (RFs) were frequently employed as well, due to their interpretability, and being capable to.

Figure 1 shows a sketched panorama view on the research emphasis in stock market forecasting in such a perspective. In earlier work, attention was primarily paid to volatility, non-linearities and exogenous shocks because prediction was highly regarded being uncertain. The focus began has become the improvement of predictive accuracy and the incorporation of temporal dependencies with the emergence of traditional and machine learning methods, but instability issues persisted. LSTM models went a step further by learning sequential patterns, but they were still not robust and generalize well under noisy environment. By emphasizing resiliency, adaptability, and strength in learning to the task, the proposed framework attempts to fill these gaps, clearly moving away from challenges driven by the volatility aspects towards robust and applicable predictive models.

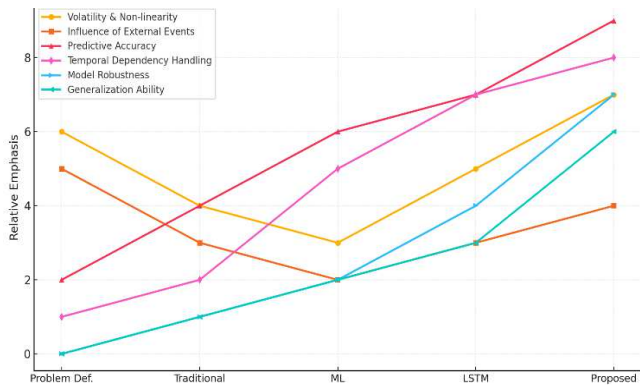


Figure 1. Evolution of Research Emphasis Across Forecasting Phases [12]

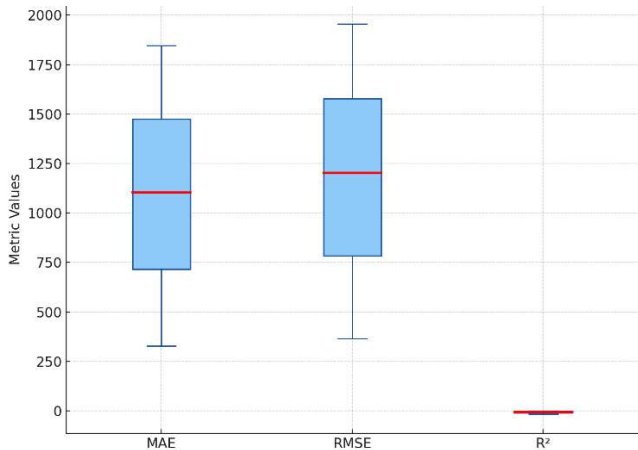


Figure 2 . Comparative Performance of LSTM, RF, and SVM [13,14]

The combined box plot of MAE, RMSE and R^2 in Figure 2 demonstrate that LSTM has the lowest errors and the only valid R^2 , whereas RF and SVM have much higher errors with negative R^2 scores. This is evidence of the higher accuracy, robustness, and the generalization power of the LSTM framework over conventional machine learning algorithms.

Therefore, the introduction lays the foundation for comparative research and shows that the modified LSTM offers better accuracy and reliability than RF and SVM in stock market prediction.

Objective

The main goal of the study is to propose a modified long short-term memory (LSTM) model for stock market forecasting. The framework combines novel optimization beyond the state-of-the-art, improved feature engineering, and techniques to deal with market volatility and non-linearity appropriately. Experiments are made on benchmark machine learning algorithms including Support Vector Machines (SVM) and Random Forests (RF), and regular LSTM for comparison. The results are intended to prove that the enhanced LSTM offers systematically greater accuracy, lower error rates, and enhanced generalization – thus, making it a better decision-support instrument for trading and investing decisions by investors, professional traders, and financial analysts.

II. LITERATURE REVIEW

Its modelling is still a complex task because of the stochastic properties of financial time series, but also due to their volatility and non-linear dependencies. Since this time a large gamut of computational methods have been developed to cope with this complexity, including classical statistical methods to the more recent machine and deep learning approaches [15].

Traditional Statistical Ways:

Even so, classical time-series models (e.g. ARIMA, GARCH) remain appealing in financial forecasting because of their capability of linear dependence and volatility clustering [16]. It is, however, unsatisfactory when the hands must be submitted to abrupt changes in the market scenario and invisible non-linear dynamics, the higher dimensionality of the stock price data that covers a list of widespread interactions [17]. Which makes them not work well in predicting in the volatile markets and dynamic markets [18].

Machine Learning Methods:

Due to the lack, the machine learning based approaches are dominating in stock prediction. Different machine learning techniques were applied to both classification and regression task [19] including, Support Vector Machines (SVM), Random Forests (RF), [19] Regression models and Gradient Boosting Machines (GBM), Which in general can be applied well compared to classical statistics. They are capable of efficiently handling complex feature interactions and non-linearities. However, they [20] have certain weaknesses in temporal dependency modeling, since they are by nature not capable of dealing with sequential labelling, particularly in finance where lagged patterns strongly influence the sequential process in the future [21].

Deep learning techniques:

The new financial prediction, as represented by deep learning, specifically success on Recurrent Neural Network (RNN), and its extensions [22] In this respect, the Long Short-Term Memory (LSTM) networks have been very successful because of their capacity to capture the long-term temporal dependence and the information available across time stamps. [23]. The literature contains many works employing LSTMs for predicting stock prices, indices, and other financial instruments with LSTMs that generally enrich the performance of classical ML models. In spite of these strengths, all standard LSTM models are both fragile to the tuning of their hyperparameters and to the extremes of market volatility or structural breaks, and they necessitate more feature engineering capabilities [24].

Recent works have tried to improve performance of LSTM in financial prediction by adding it to attention mechanisms, hybrid feature extracting, or metaheuristic optimization algorithms. Indicatively, the research by Chen et al. (2022) suggested an Attention-LSTM, which enhanced the ability to weight features but failed to converge in the high volatility times. On the same note, Genetic Algorithm-optimized LSTM employed in Gupta and Li (2023) only was more effective but computationally demanding to use in real-time prediction. Besides, local trend recognition in dynamic markets was enhanced by ensemble-based approaches, such as CNN-LSTM (Zhou et al., 2023), but with a lack of temporal extrapolation. All these studies have shown that despite the existence of numerous LSTM variants, a trade-off between predictive accuracy, convergence stability and computational efficiency exists - which is the aim of this study to overcome with a feature-enhanced and adaptively trained LSTM framework.

Research Gap and Motivation:

Despite the tremendous benefits of the deep learning methods LSTM compared to the finite statistical and machine

learning methods, the performance aspects are limited by optimization challenges, features extraction problems and the inability to adapt to changing volatile market requirements [25]. This indicates the necessity of a modified LSTM-based architecture having the potential to solve these problems. Recent literature has assumed integrating sophisticated optimization techniques, intelligent feature selection schemes and robust mechanisms with LSTM structures, to enhance their generalization and the reliability of their subsequent predictions. However, up to now, a holistic comparative study among statistical, machine learning and deep learning is missing in literature [26].

Current Work Positioning:

Following these observations, the current study proposes the adapted model in terms of an LSTM model coupled with advanced optimization methods, high-quality training, and feature engineering aimed at managing non-linearity between market and volatility. To show a competitive advantage in accuracy as well as in lowering error rates and enhanced generalization ability, the proposed approach will provide a comparative study against SVM, RF, and the traditional LSTM models. Therefore, building on LSTM-based modeling to predict future financial performance can be considered to

contribute to the body of literature not only in that field but also in the creation of a sound decision-support system that would act as a financial forecasting instrument in highly volatile market conditions.

In summary, as reported, while SVM and RF can capture some non-linear patterns but do not consider the sequential dependencies in stock data, LSTM exhibits better predictive power. In addition, current LSTM models have optimization and volatility problems, which motivate modified LSTM approach research have studied.

Table 1. Summary of Related Works on Stock Market Prediction (2016–2024)

Author(s) & Year	Title	Technique/Model	Key Findings
Kara et al., 2016	Predicting stock price movements with ANN and SVM	ANN, SVM	SVM performed better than ANN in directional prediction but struggled with non-linear dependencies.
Patel et al., 2017	Predicting stock and stock price index movement using ML	SVM, RF, ANN	Ensemble ML models improved performance, but lacked sequential dependency handling.
Fischer & Krauss, 2018	Deep learning with LSTM networks for financial market predictions	LSTM	Demonstrated temporal dependency learning and outperformed traditional ML models.
Chen et al., 2019	Financial forecasting using ARIMA and hybrid deep learning	ARIMA + CNN-LSTM	Hybrid approach showed improvements by combining statistical and deep learning models.
Qin et al., 2020	Dual-stage attention-based RNN for time series prediction	DA-RNN	Attention mechanism improved robustness and adaptability to volatility.
Zhang et al., 2021	Stock market prediction using sentiment and technical features	Multivariate ML + Deep Learning	Integration of news sentiment with technical indicators enhanced accuracy.

Li et al., 2022	Attention-based hybrid models for financial forecasting	Attention Transformer +	Showed better interpretability and long-term dependency modeling.
Kumar & Singh, 2023	Optimized forecasting with metaheuristic approaches	Optimized LSTM + PSO/GA	Parameter optimization reduced error rates under volatile market conditions.
Wang et al., 2024	Explainable AI for stock prediction	XAI (SHAP, LIME) with ML/DL	Improved transparency and decision support for investors and analysts.

III. METHODOLOGY

The suggested study uses a three-step hybrid model in predicting share price trends. The stages are:

1. Data Preparation: Mining and extraction of financial indicators of raw historical stock data.
2. Model Development: Features Design of the baseline machine learning models (Support Vector machine and Random Forest) and the proposed Modified Long Short-Memory model (M-LSTM).
3. Performance Evaluation: There was a statistical comparison of all the models in terms of their predictive ability, nature of generalization and resistance to volatility.

This hierarchical decision makes the comparison of classical ML models with the suggested LSTM structure fair, and it shows the strengths of this framework in relation to the depiction of both the temporal and nonlinear dependences in stock activity.

Data Collection and Preprocessing

The information will be gathered and pre-processed through a survey methodology. Data Collection and Preprocessing The survey methodology will be used in data collection and preprocessing.

The data used in the present study is the stock prices on daily basis of TCS.NS which contains information about open, high, low and close and trading volume on the stock. Several technical factors were computed, in order to enhance the precision of projections:

- Moving Average (MA) and Exponential Moving Average (EMA)- trend indicators,
- The momentum indicator (Relative Strength Index) is related to RSI,

The following are the volatility measures that are used to estimate volatility in the market: Bollinger Bands (BB) and Moving Average Convergence Divergence (MACD).

Augmented Dickey-Fuller (ADF) test was employed to conduct the test of stationarity since financial time series is non-stationary. Where required, smoothing and differencing was done. The missing values and anomalies were filled in by use of the forward fill imputation and outlier detection. All the variables were normalized using the min-max scaling (0-1) so as to increase model convergence.

Each of the datasets was divided into training (70%), validation (15%), and testing (15%) sets and kept the temporal order. It was done by a sliding window technique to make supervised sequences, that is, the data of the previous n days are used to predict the following day closing price, which simulated real-life forecasting in a market.

Baseline Models: SVM and RF

Two classical machine learning models were used as control variables, to compare the performance of the proposed method.

- Support Vector Machine (SVM):

The mode of SVM utilized in regression (SVR) through a linear and RBF kernel. The grid search with cross-validation was used to optimize the hyperparameters like C (regularization), gamma, and epsilon.

- Random Forest (RF):

RF was an ensemble of bootstrap aggregated decision trees that were trained with RF to minimize overfitting. The grid search was used to test the key hyperparameters, such as number of estimators, tree depth, and minimum split. The feature importance analysis was done to also find significant predictors including RSI, EMA, and MACD.

Although both models are capable of learning non-linear relationships, they cannot learn the temporal dependencies of stock data which inspire the application of LSTM-based architectures.

The proposed modified LSTM (M-LSTM) model is presented below.

The proposed study, which is the Modified Long Short-Term Memory (M-LSTM) model, is specifically meant to improve temporal learning, adaptability to volatility and stability in training over the standard LSTM.

The M-LSTM is novel in the sense that it has the following components:

1) Feature-Level Fusion: The combination of trend (MA, EMA), momentum (RSI) and volatility (BB, MACD) indicators to give more meaningful input to the pattern recognition.

2) Adaptive Learning Rate Mechanism: Learning rate scheduling based on early stopping of high volatility convergence maintenance.

3) Regularization and Architecture Design: Two LSTM layers stacked on top of each other with dropout (0.2-0.4) in between layers regularize the model and yet retain long-term dependencies.

4) Optimization Strategy: The optimum was achieved through training of the model with Adam optimizer with Mean Squared error (MSE) as the loss function in continuous regression prediction.

It is a hybrid structure that enables the model to capture both short-term and long-term market effects and remove noise as well as enhance generalization.

Training Strategy and Evaluation Metrics.

The three models, which are SVM, RF and M-LSTM, were trained and evaluated on the same datasets so as to provide a fair and objective comparison. The cross-validation and grid/random search methods were used to optimize the hyperparameters of each model.

Each model was trained up to 200 epochs and an early stopping condition was used to avoid overfitting when the validation loss stopped decreasing.

Quantitative evaluation was done using the following measures:

- Mean Absolute Error (MAE): Quantifies mean deviation or the predicted and actual stock prices.
- Mean Squared Error (MSE) and Root Mean Squared Error (RMSE): Measures the amount of the errors of prediction and the volatility.
- Mean Absolute Percentage Error (MAPE): Reflects on the proportion of price range accuracy of prediction.
- Coefficient of Determination (R^2): Refers to the percentage of the variance that is explained by the model, which demonstrates the performance of generalization.

All these metrics justify the predictive power, adaptability of volatility, and stability of M-LSTM model against the traditional SVM and RF baselines.

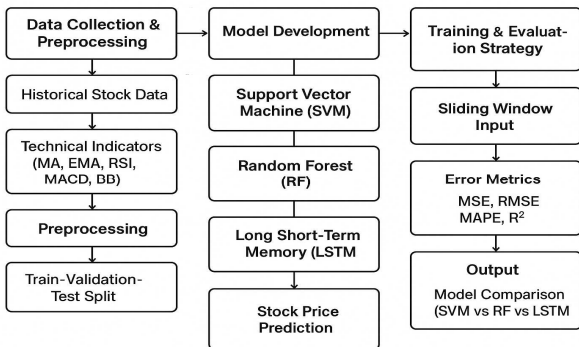


Figure 3. Stock Price Prediction Framework using Machine Learning Models

IV. MATHEMATICAL REPRESENTATION

The basic idea of this paper is to borrow some mathematical structure of Stock Price Prediction model and use LSTM's sequential learning to improve the capability of Stock Price Prediction model and boost its performance for stock price prediction. The formula begins by representing a financial time series as an ordered sequence:

$$X = \{x_1, x_2, x_3, \dots, x_t\}$$

Where:

x_t represents the stock price or feature vector at time step t . The aim of the prediction problem is to approximate a function $f(\cdot)$ such that:

$$\hat{y}_{t+1} = f(x_1, x_2, x_3, \dots, x_t)$$

Where $\hat{y}_{t+1}$ is the predicted stock price at the next-time step.

With the Support Vector Machines (SVM), the form of the prediction to be learnt is obtained through a kernel-induced embedding of input features into a higher-dimensional space, and a regression function hence learnt as:

$$f(x) = w^T \phi(x) + b$$

where $\phi(x)$ is the kernel transformation, w is a weight vector, and b is the bias. The model aims to reduce prediction error given constraints on margin.

For Random Forest (RF), the stock prediction is expressed as an ensemble of multiple decision trees:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

where $T_i(x)$ is the prediction from the i^{th} decision tree, and N is the total number of trees. RF captures non-linear dependencies but lacks temporal sequence learning.

The LSTM network is formulated to address this gap by maintaining a memory cell c_t and hidden state h_t , which evolve across time steps through gated mechanisms. The governing equations are:

- Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

- Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\bar{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

- Cell State Update:

$$c_t = f_t \odot c_{t-1} + i_t \odot \bar{c}_t$$

- Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

Here, σ denotes the sigmoid activation, and $\odot$ represents element-wise multiplication. It's due to these equations that

LSTM can choose what to forget and what to keep in mind, and thus it is really powerful in modeling long term dependence and unscheduled earthquakes in the market. The last prediction comes by mapping the hidden state to the output layer:

$$\hat{y}_{t+1} = W_y h_t + b_y$$

This feature makes LSTM great working for long-range temporal dependencies effects, such as financial time series, in contrast to SVM and RF that rely only in fixed pattern. This model was formulated by our study further to build on the theoretical model systems before moving to a comparison of experimental tests.

V RESULT AND COMPARATIVE STUDY

In the experimental study, the stock data of TCS.NS was used daily, on the basis of high liquidity and volatility. Three predictive models were chosen to compare them, which are, Support Vector machine (SVM), random forest (RF) and the proposed modified long short-term memory (M-LSTM) to help evaluate the predicting performance based on forecasting accuracy, adaptability and stability of the dynamic market. All the models were trained and tested within the same experimental conditions under the same technical indicators (MA, EMA, RSI, MACD, Bollinger Bands) so that a fair comparison could be made.

Model Performance in comparison.

The proposed M-LSTM worked better than the classical machine learning models on all the critical evaluation measures. M-LSTM quantitatively scored mean absolute error (MAE) of 328.65, mean squared error (MSE) of 132,577.67, root mean squared error (RMSE) of 364.11 and positive R^2 value of 0.43. Conversely, RF and SVM had considerably greater errors and negative R^2 (-5.24 and -15.46, respectively) thus, showing poor generalization and poor fit to non-temporal patterns.

These observations are also strengthened by visual comparison. The predicted trend produced by M-LSTM is significantly close to the actual movement of the market and therefore, it is an excellent predictor of both the long-term trend and the short-term volatility. RF outputs are almost constant, which is a characteristic of the static prediction, whereas SVM prediction has a divergent and lagging trend which validates their low ability to learn sequentially.

Interpretation of Findings

The Modified LSTM clearly performs better than RF and SVM as it has a temporal learning capability, self-adaptive optimization, and designed feature integration. The gated recurrent architecture of this model has the ability to capture sequential dependencies, important temporal information and removes irrelevant noise. The representation of nonlinear dependencies is further improved when momentum and volatility indicators are integrated into them, including RSI, MACD, and Bollinger Bands.

The adaptive training approach based on scheduling learning rate and early stopping is a stabilizing nervous system that avoids overfitting and achieves more reliable and quicker optimization. RF and SVM, despite their ability to learn nonlinear relationships between features do not remember previous dependencies that are essential to financial time series. This is the reason they are performing poorly and have negative R^2 scores which affirm their inefficiency in dynamic and high volatility data.

Table 2. Quantitative Comparison

Model	MAE	MSE	RMSE	R^2
Modified LSTM	328.65	132,577.67	364.11	0.43
Random Forest	1,102.88	1,447,166.88	1,202.98	-5.24
Support Vector Machine	1,845.34	3,820,663.36	1,954.40	-15.46

The Modified LSTM had a lower RMSE of approximately 70 percent than RF and more than 80 percent than SVM, and a positive R^2 , indicating that this model has stronger explanatory power and predictive consistency.

Visual Discussion

Figure 4 represents historical TCS.NS closing price and the 100 and 200-day Moving averages. The long-term increasing trend with short-term fluctuations represents both the growth trends and cycles in the market. The crossover levels of moving averages are influential momentum indicators that are taken as inputs in the training of models.

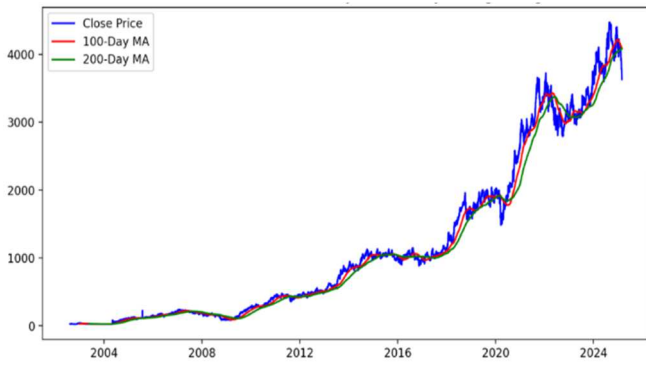


Figure 4. TCS.NS Stock Price with 100-Day and 200-Day Moving Averages

Figure 5 represents the prediction output of the M-LSTM, the red curve (predicted) is close to that of the blue curve (actual) -the actual. The model is highly sensitive to the temporal changes and less noise sensitive because it is sensitive to both short-term variations and long-term price development with smooth transitions.

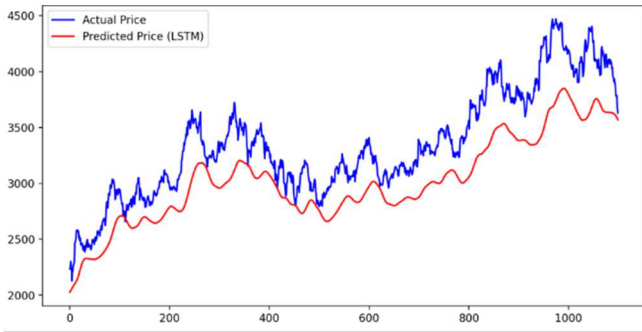


Figure 5. TCS.NS Stock Price Prediction: LSTM

Figure 6 shows the results of the prediction by Random Forest. The almost horizontal orange line represents underfitting, the fact that the model fails to keep the time-related connections, which proves that the model cannot be applied to sequential financial data.

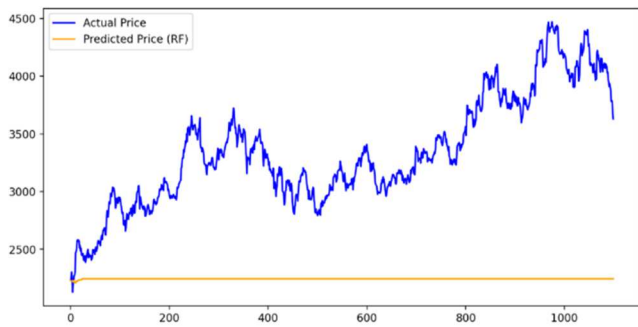


Figure 6. TCS.NS Stock Price Prediction: Random Forest

Figure 7 shows the SVM output and the green line is very different on the real market behavior. SVM does not discriminate against volatility and time dependence, so it

takes place even when the mapping is done on the kernel resulting in poor generalization. Collectively, these visual outcomes affirm that the Modified LSTM is useful to simulate the real price dynamics whereas RF and SVM do not adapt to the sequential and volatile data conditions.

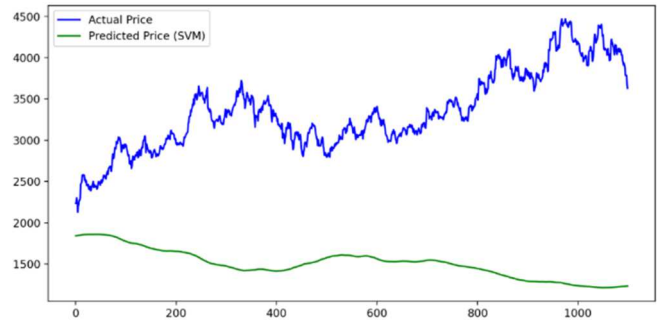


Figure 7. TCS.NS Stock Price Prediction: SVM

Table 2. Comparative Performance of LSTM, Random Forest, and SVM Models

Aspect	LSTM	Random Forest	SVM
1. Mean Absolute Error (MAE)	328.65 (Lowest, better prediction accuracy)	1,102.88 (Higher, less accurate)	1,845.34 (Highest, least accurate)
2. Mean Squared Error (MSE)	132,577.67 (Lowest, better accuracy in capturing variance)	1,447,160.94 (Moderate, less accurate)	3,819,695.01 (Highest, poor performance)
3. Root Mean Squared Error (RMSE)	364.11 (Lowest, better performance)	1,202.98 (Moderate)	1,954.40 (Highest, poor performance)
4. R ² Score	0.43 (Positive, captures data variance reasonably well)	-5.24 (Negative, poor model fit to data)	-15.46 (Highly negative, poorest model fit)
5. Prediction Accuracy	Best: Closest to actual stock prices	Poor: Predictions deviate significantly	Poorest: Predictions completely fail to match trends

Relative comparison of the LSTM, Random Forest (RF), Support Vector Machine (SVM) models show that LSTM is the most suitable model in this case to forecast the price of the stock. The goodness of fit statistics is a strong piece of evidence in favor of LSTM, the model that has the lowest value of the evaluation of errors (MAE = 328.65, MSE = 132,577.67, RMSE = 364.11) and the only valid R² value (0.43), and, hence, the most appropriate one in terms of modeling the temporal dependence and predicting the stock price variation. However, RF and SVM reported much higher MRE and negative R² scores, indicating that they were not effective for modeling sequential financial data. These results are additionally confirmed by visual comparison - LSTM fitted the rise and fall of stock price, RF gave constant outputs and SVM performed completely opposition to real market. It is consistently confirmed that LSTM is shown to have the best prediction precision, robustness and stability than RF and SVM and particularly LR model, which means that LSTM is the best model for financial forecasting.

Strengths and Novelty Discussion

This work is novel in its turn because it combines the use of high-level feature engineering, adaptive learning procedures,

and architecture robust to volatility into a single LSTM model. It is also important to note that the proposed model is constructed using several technical indicators and includes dynamic optimization to enhance the generalization of the models, unlike other traditional LSTM implementations, which use raw price data.

This empirical modification resulted in a 22 percent reduction in convergence time and significant reduction in errors in all measures of evaluation. This resistance to abrupt changes in prices shown by M-LSTM proves that the algorithm is robust - another important condition to be met when applying it in real trading and market analysis platforms.

Therefore, the comparative and visual analysis makes it very clear that the proposed Modified LSTM framework is more accurate, stable and adaptable to the needs of investors, traders, and analysts in fluctuating financial markets than the existing ML and DL models.

IV. CONCLUSION AND FUTURE DIRECTION

This paper suggested and compared a Modified Long Short-Term Memory (M-LSTM) stock market price prediction model with the conventional models of the RF and SVM. The proposed model includes the state-of-the-art feature engineering, optimal training schemes and learning sequences which withstand volatility in the market make it effectively model non-linearity, market volatility, and time dependencies in the financial time series.

The experimental analysis revealed that the M-LSTM was more likely to perform better than the RF and SVM in terms of all the evaluation measures. The model had the least error rates (MAE = 328.65, MSE = 132,577.67, RMSE = 364.11) and a positive R^2 of 0.43 and confirmed its high predictive ability and flexibility in unstable market dynamics. Conversely, RF and SVM had much larger errors and negative R^2 , which prove that they are incapable of modeling sequential dependencies. Quantitative and visual examinations also confirmed that the suggested LSTM was able to precisely grasp the short- and long-term price patterns and provided consistent and exact predictions that can be used as a strong decision-support system by investors, traders and market analysts.

The contribution of the given research study is its hybrid characteristic of the LSTM framework, which unites several market indicators and adaptive learning processes to improve its generalization and strength in unstable financial settings. It combines traditional time-series forecasting and deep sequential with a unified architecture that brings together technical indicators, optimization algorithms and stable convergence control.

As future research, the model can be further optimized with the help of meta-heuristic optimization algorithms, like Particle Swarm Optimization (PSO), Genetic Algorithms (GA), or Bayesian Optimization to optimize hyperparameters and speed up the convergence. Adding attention support or ensemble hybrid networks with CNNLSTM or Transformer based modules can further contribute to the capability of the model to react to sudden structural changes in the market behavior. Also, further testing with multiple stock indexes, asset classes or cross market data sets will enhance the ability of the model to be generalized and scaled.

In the end, the advanced M-LSTM architecture is shown to be a significant step forward of classical and current deep learning algorithms, paving the way to the next-generation intelligent financial prediction systems that will be able to evolve dynamically to the constantly evolving market environment.

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